

Green building adaptation in hot-humid climates: assessment of coconut and corn husk fiber composite bricks as energy-efficient building envelopes

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Purpose

This study has assessed the thermal performance of locally fabricated bio-based building envelopes made of coconut and corn husk composite bricks to reduce building wall heat transmission load and energy consumption towards green building adaptation.

Design/methodology/approach

Samples of coconut fiber (coir) and corn husk fiber bricks were fabricated and tested for their thermophysical properties using the Transient Plane Source (TPS) 2500s instrument. A simulation was conducted using Dynamic Energy Response of Building - Lunds Tekniska Hogskola (DEROB-LTH) to determine indoor temperature variation over 24 h. The time lag and decrement factor, two important parameters in evaluating building envelopes, were also determined.

Findings

The time lag of the bio-based composite building envelope was found to be in the range of 4.2–4.6 h for 100 mm thickness block and 10.64–11.5 h for 200 mm thickness block. The decrement factor was also determined to be in the range of 0.87–0.88. The bio-based composite building envelopes were able to maintain the indoor temperature of the model from 25.4 to 27.4 °C, providing a closely stable indoor thermal comfort despite varying outdoor temperatures. The temperature variation in 24 h, was very stable for about 8 h before a degree increment, providing a comfortable indoor temperature for occupants and the need not to rely on air conditions and other mechanical forms of cooling. Potential energy savings also peaked at 529.14 kWh per year.

Practical implications

The findings of this study present opportunities to building developers and engineers in terms of selecting vernacular materials for building envelopes towards green building adaptation, energy savings, reduced construction costs and job creation.

Originality/value

This study presents for the first time, time lag and decrement factor for bio-based composite building envelopes for green building adaptation in hot climates, as found in Ghana.

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